ENTERPRISE PROJECT SPECIFICATION

VANI COLLECTION

High-Fashion Artisanal E-Commerce Platform & Full-Stack Atelier Management Suite

VERSION	ARCHITECTURE	DATABASE	DATE
Production 2.4	**Next.js 16 + Express API**	**MongoDB Atlas Enterprise**	**October 2026**

CORE TECHNOLOGY STACK

Next.js 16 (App Router) · React 19 · Turbopack · Tailwind CSS v4 · Framer Motion 13 · Lenis Smooth Scroll · Node.js + Express REST API · MongoDB Mongoose ODM · Cloudinary CDN · Razorpay Payment Ready · TypeScript Strict

EXECUTIVE SUMMARY

Vani Collection is an end-to-end custom digital boutique crafted for luxury handwoven ethnic couture (Pure Mul Cotton, Chanderi, Bagru prints, and festive heirlooms). Engineered without template overhead or high recurring Shopify plugin costs, it provides bespoke editorial storytelling, an interactive shoppable Lookbook with coordinate hotspots, Instagram-style category story reels, seamless bag/checkout mechanics, and a non-technical staff Admin Console for 100% self-managed catalogue and homepage operations.

Prepared for: Vani Collection Stakeholders

Confidential & Proprietary

1. Storefront Experience & Brand Identity

FRONT-OF-HOUSE

The customer-facing portal is designed with a high-fashion Jaipur atelier aesthetic, utilizing warm ivory textures (`#faf7f2`), rich deep maroon/burgundy accents (`#881337`), and vintage gold leaf embellishments (`#dfc28c`, `#b8935a`).

✨ Dual-Mode Hero Experience

Allows visitors to toggle instantly between:

- **Auto-rotating High-Res Slideshow:** Editorial campaigns with custom eyebrow tags, headlines, focal points, and CTAs.
- **Full-Motion Video Cinema:** Embedded artisanal weaving video showcase with pause/play sound controls.

🌿 Kinetic Smooth Scroll (Lenis Engine)

Integrated with Lenis inertia physics for buttery 60fps scrolling across mobile and desktop. Smart auto-pause disables scroll momentum when search overlays, cart drawer, or modal reels are triggered to eliminate layout jitter.

📱 Native Mobile App Navigation

Mobile users get a sticky bottom navigation bar with direct thumb shortcuts to Storefront, Category Explorer, Wishlist counter, and Instant Bag drawer with badge notifications.

📢 Continuous Announcement Marquee

Top banner features an infinite smooth CSS marquee ticker highlighting coupon promotions ("VANIFESTIVE15"), worldwide shipping alerts, and Jaipur atelier authenticity guarantees.

2. Interactive Shoppable Editorial Lookbook

SIGNATURE FEATURE

⭐ Multi-Lookbook Editorial Switcher with Ambient Luxury Canvas

A custom editorial styling experience that turns fashion photography into direct purchase pathways:

- **Multi-Lookbook Support:** Admin can publish multiple independent lookbooks. Visitors switch between them smoothly via Look 1, Look 2 pills and desktop next/prev arrow buttons.
- **Luxury Ambient Theater Frame:** Fixes the vertical model photo cropping issue. Model is displayed in full natural 3:4 portrait height (head to toe uncropped) while outer widescreen space is illuminated by a soft, glowing ambient blurred reflection of the outfit.
- **Shoppable Hotspot Pulse Pins (+):** Gold pins pulse subtly over the muse's outfit. Clicking a pin opens a floating glassmorphic tooltip card.
- **Dynamic Catalogue Product Linking:** Pin tooltips display the linked product's real image thumbnail, title, category/fabric tag, formatted INR price, Quick View modal trigger, and instant "+ Bag" button.
- **Dedicated Navbar Direct Scroll:** Clicking "LOOKBOOK" in the header navigation or dropdown automatically smooth-scrolls straight to `#lookbook`.

3. Instagram-Style Category Stories & Video Reels

ENGAGEMENT ENGINE

🎬 Tap-to-Watch Story Tiles

Instagram-style circular and card tiles showcasing artisanal categories (Pure Mul Cotton, Festive Chanderi, Bagru Handblock, Royal Anarkalis). Unviewed stories feature an animated gradient ring.

📱 Fullscreen Reel Player Modal

Fullscreen immersive modal with auto-advancing progress timers, sound toggle, swipe/tap navigation between stories, and a direct "Shop this Collection" button linked to active category catalogue.

4. Product Catalogue & Smart Discovery

SHOPPING MECHANICS

🛍️ Living Product Cards

- **Interactive Image Flip:** Hovering swaps to alternate model angles / fabric close-up.
- **Quick Size Pill Selector:** Pick XS, S, M, L, XL, XXL directly on the card without opening product page.
- **Instant Heart Wishlist Toggle:** 1-tap save/remove with animated counter.
- **Dynamic Badges:** "Bestseller", "New Launch", "Pure Mul", "Festive Luxe".

🔍 Quick View Modal & Fast Search

- **Instant Quick View:** Opens full product specifications, image gallery, fabric details, size chart, and add-to-bag without leaving the homepage.
- **Instant Debounced Search:** Header search overlay queries titles, fabrics, tags, and categories in real-time with zero page reload.

CATALOGUE FEATURE	IMPLEMENTATION	BUSINESS ADVANTAGE	STATUS
Category Filtering	Mul Cotton, Festive, Co-ord Sets, Anarkalis, Sarees, Dupattas, Under ₹1,999	Allows customers to find desired style in 1 click	ACTIVE
Dynamic Variant Matrix	Multi-size SKUs (XS, S, M, L, XL, XXL) with independent inventory units	Prevents overselling out-of-stock sizes	ACTIVE
Fabric & Craft Details	Detailed Jaipur weaving notes, 100-count Mul certifications, care guidelines	Builds consumer trust for premium handcrafted items	ACTIVE
Price Range Slicing	Budget filter ("Under ₹1,499", "₹1,500 - ₹1,999", "Premium Collection")	Accommodates impulse buyers and luxury bridal shoppers	ACTIVE

5. Shopping Bag, Coupons & Checkout Flow

CONVERSION ARCHITECTURE

🛒 Slide-Out Shopping Bag Drawer

Persistent drawer accessible from any page:

- **Free Shipping Bar:** Live dynamic progress meter indicating "Add ₹X more for Free Express Delivery".
- **Quantity Increment/Decrement:** In-cart quantity editing with live subtotal calculation.
- **Size & Variant Display:** Explicit breakdown of selected dress sizes.
- **Client-Side Persistence:** Cart automatically preserved in browser storage across sessions.

💳 Secure Multi-Gateway Checkout

Frictionless multi-step checkout:

- **Promo Code Engine:** Validates coupons (flat INR off / percentage discount) with instant deduction.
- **Delivery Address Book:** Pin code auto-detection, phone number, and delivery instructions.
- **Payment Modes:** Pre-integrated for UPI / Cards / Netbanking (Razorpay) + Cash on Delivery (COD).
- **Order Confirmation Screen:** Detailed order receipt with tracking ID.

6. Customer Portal & Post-Purchase Services

RETENTION & TRUST

📦 Live Order Tracking

Milestone tracker displaying status steps: Order Placed → Quality Check Jaipur → Dispatched → In Transit → Out for Delivery → Delivered, with courier tracking link.

⭐ Verified Reviews & Customer Rewards

Customers can post photo reviews, rate products 1-5 stars, and earn loyalty points redeemable as wallet discounts on subsequent orders.

7. Enterprise Admin & Content Management Console

BACK-OF-HOUSE CONTROL

Secured under `/admin` with role-based JWT authentication (Super Admin, Catalogue Manager, Fulfillment Staff). Built with dedicated non-technical forms—**zero raw JSON or code editing required by staff.**

📊 Executive Dashboard (`/admin`)

Real-time analytics including Total Gross Revenue (GMV), Total Completed Orders, Active Customer Count, Low Stock Warning Badges, and recent activity logs.

👗 Product Catalogue Manager (`/admin/products`)

Add/edit products with title, category, fabric type, regular price, sale price, SKU variants, sizes array, primary & hover images, video reels, and stock counts.

🎨 Visual Lookbook & Hotspot CMS

Edit editorial lookbooks visually:

- **Product Dropdown Selector:** Pick any store product from dropdown to automatically fill Name, Price, and Tag.
- **Slider Sliders:** Intuitive 0–100% Top and Left sliders to place pins.
- **Layout & Focal Controls:** Select Luxury Portrait Frame vs Widescreen Banner, and focus on Head/Face vs Center.

💬 Customer Testimonials & Reviews CMS

Dedicated form for customer reviews with Customer Name, City, Rating (1–5 Stars), Headline, Comments, Product Name, and "Mark as Verified Buyer ✓" badge toggle.

🚚 Order Management & Fulfillment (`/admin/orders`)

Track customer orders, update delivery status (Pending, Processing, Shipped, Delivered, Cancelled), add courier tracking numbers, and view invoice receipts.

🎟️ Coupon & Promotional Discounts (`/admin/coupons`)

Create coupon codes with percentage or flat discounts, minimum order value thresholds, expiry dates, and maximum redemption limits.

📦 Inventory & Stock Alerts (`/admin/inventory`)

Real-time stock monitoring with low-inventory threshold triggers. Prevents customer checkout once stock hits zero.

⚙️ Navbar & Header Mega Menu CMS

Customize announcement marquee messages, active header navigation links, dropdown category columns, brand logo, and color palettes.

8. Performance, SEO & Security Highlights

TECHNICAL STANDARDS

STANDARD	IMPLEMENTATION DETAILS	AUDIT METRIC
Core Web Vitals	Next.js 16 SSR + Static page prerendering (51/51 routes generated statically)	Sub-second LCP & 95+ Mobile Score
Search Engine Optimization	Dynamic OpenGraph images, automated `sitemap.xml`, `robots.txt`, and FAQPage JSON-LD schema	100% Rich Google Snippet compliance
Data Security	HTTP-only cookies, bcrypt password hashing, JWT stateless authorization, CORS origins	Enterprise Grade Security
Responsive Layout	Tailwind v4 fluid breakpoints from 360px mobile to 1920px 4K ultrawide monitors	Pixel-perfect multi-device fidelity